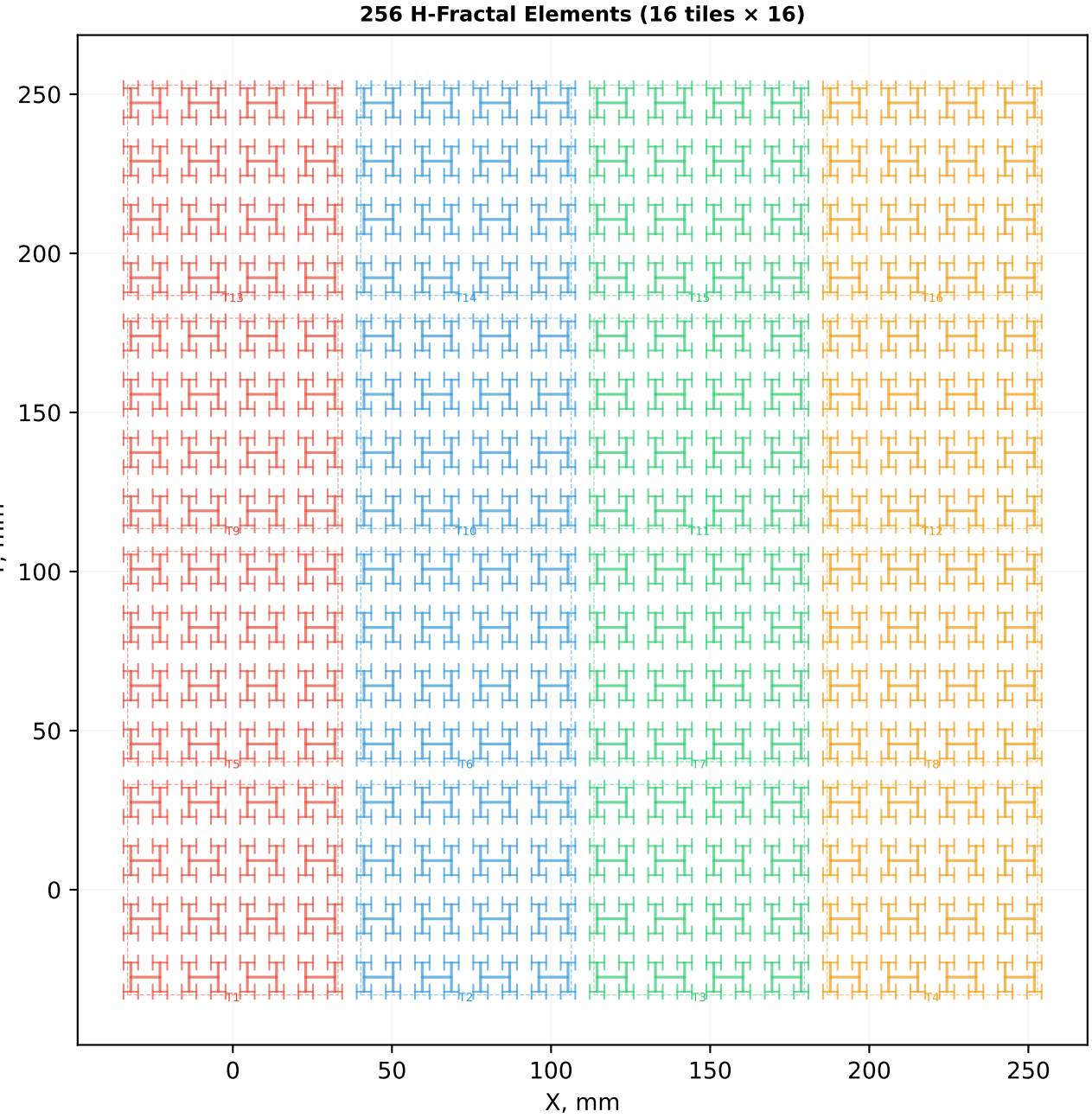
256-Element Tiled Phased Array — Simulation Report

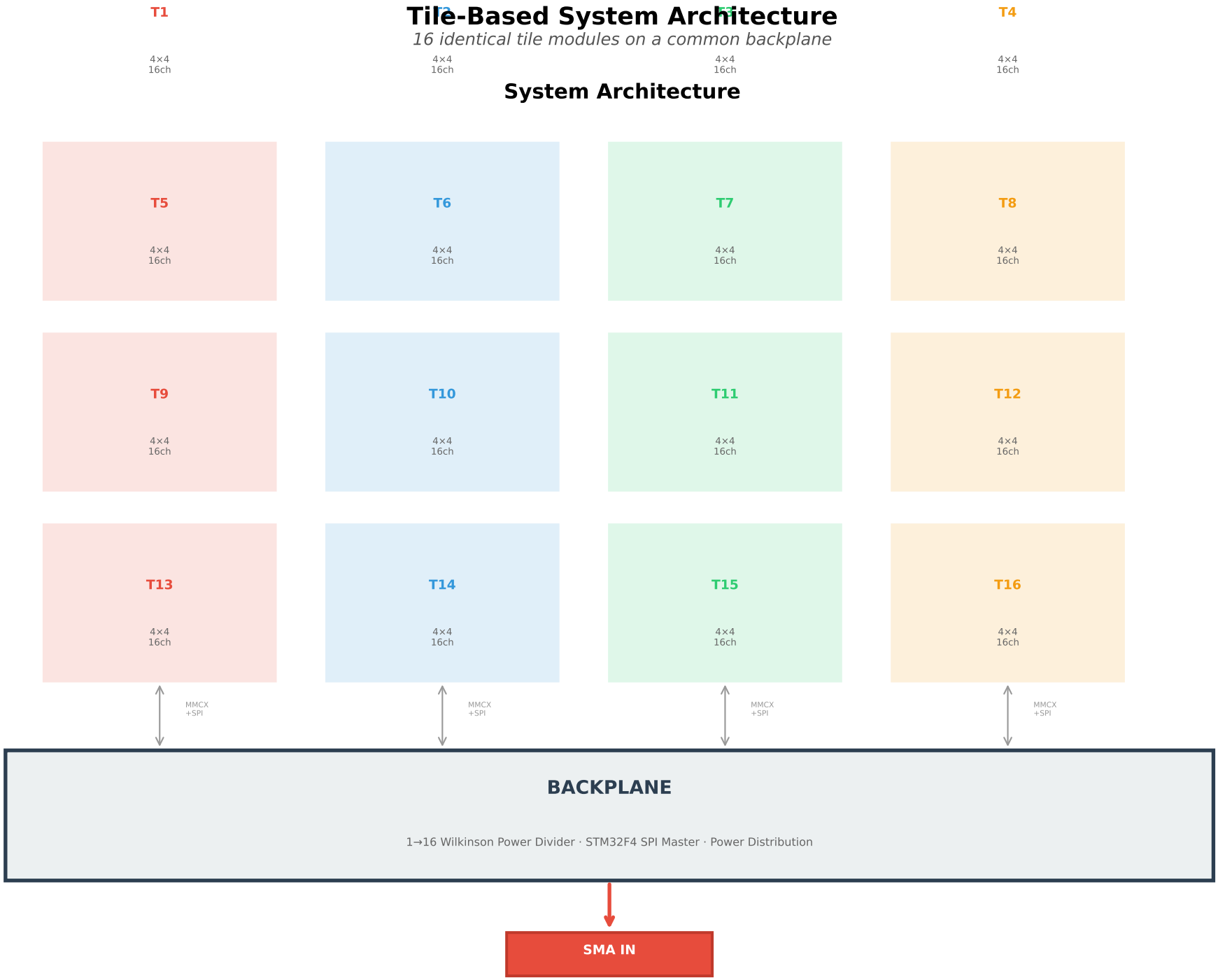
16 H-Fractal Tiles · 64× ADAR1000 · 10.5 GHz Radar



System:	AERIS-10N 256-Element Radar
Topology:	H-Fractal Tiled Array
Elements:	256 (16×16)
Tiles:	16 (4×4 grid)
Channels:	256 (64× ADAR1000)
Frequency:	10.5 GHz
λ_0:	28.55 mm
Spacing:	18.3 mm (0.64λ)
Aperture:	275×275 mm
Tile size:	73×110 mm
Tile pitch:	73.27 mm
Backplane:	300×360 mm
MCU:	STM32F4 (SPI)
Substrate:	RO4350B

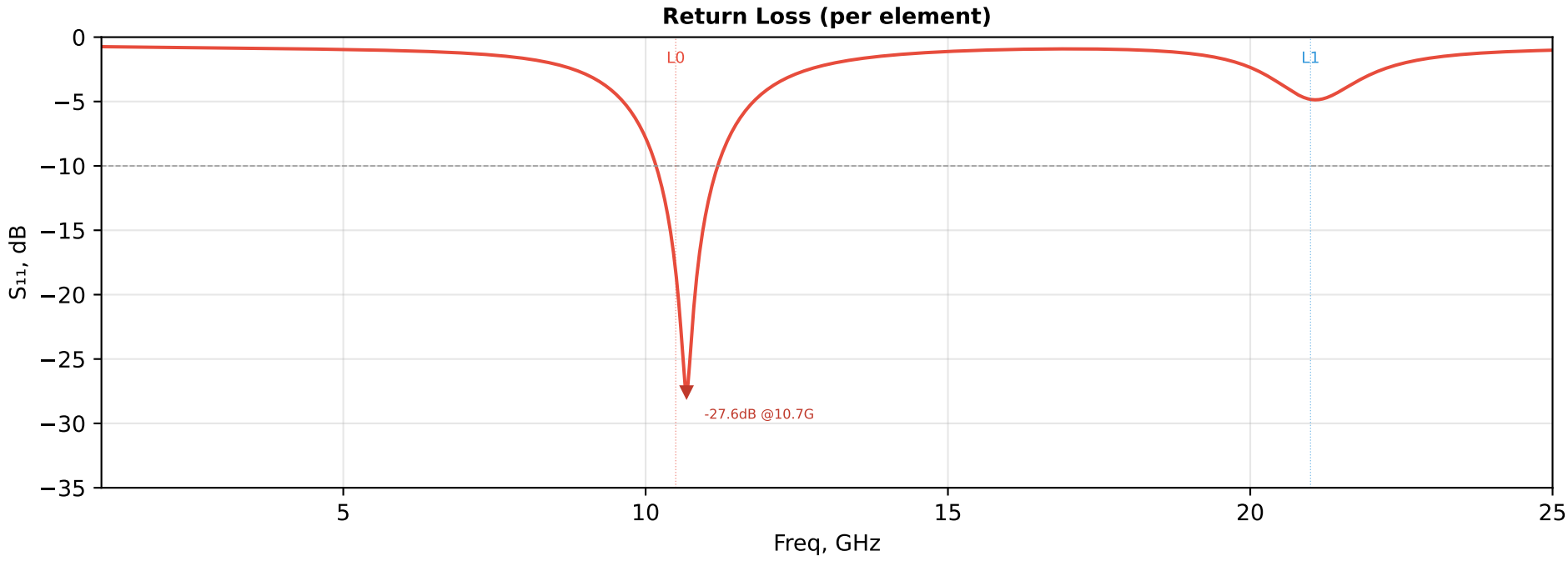
Tile-Based System Architecture

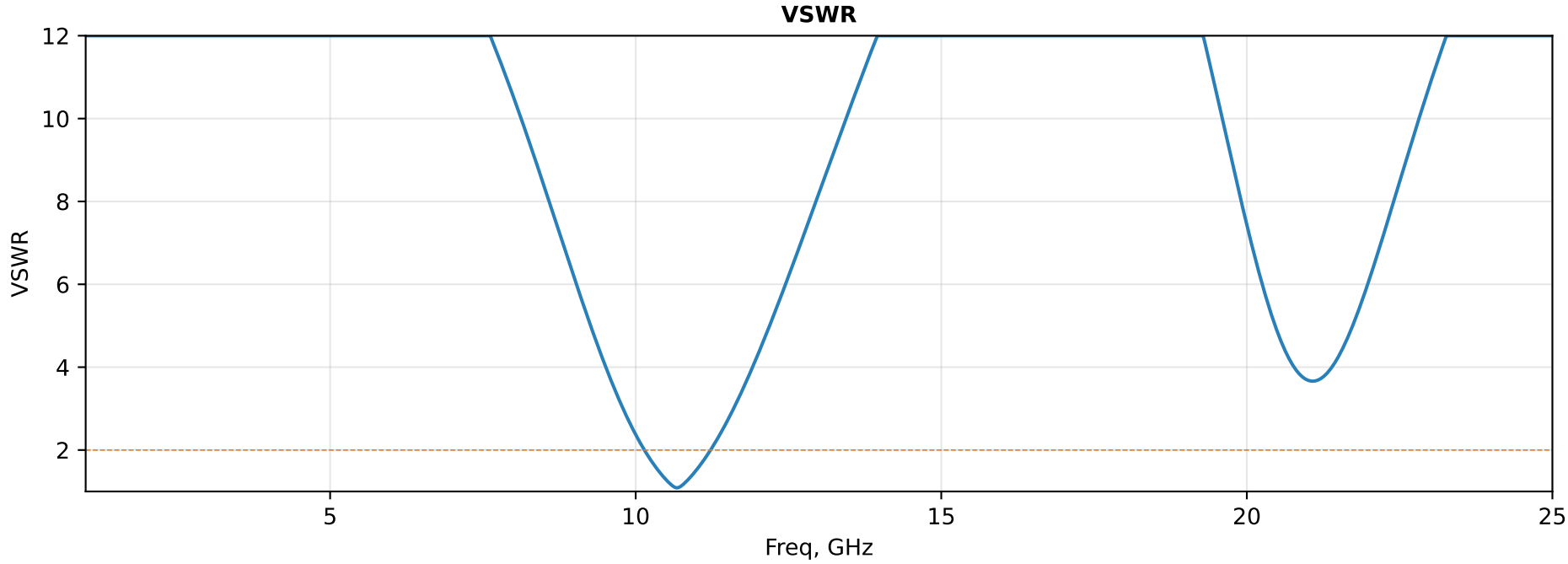
16 identical tile modules on a common backplane



S_{11} and VSWR (per element)

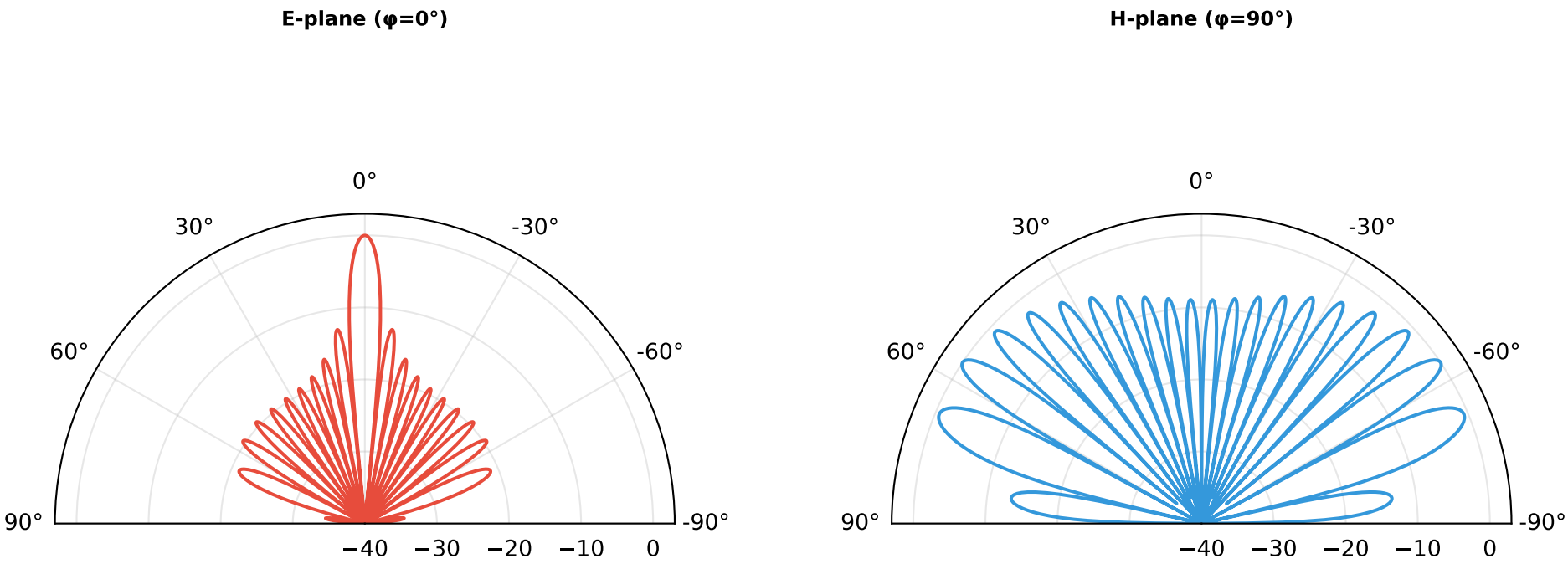
Identical for all 256 elements — multi-band fractal resonances





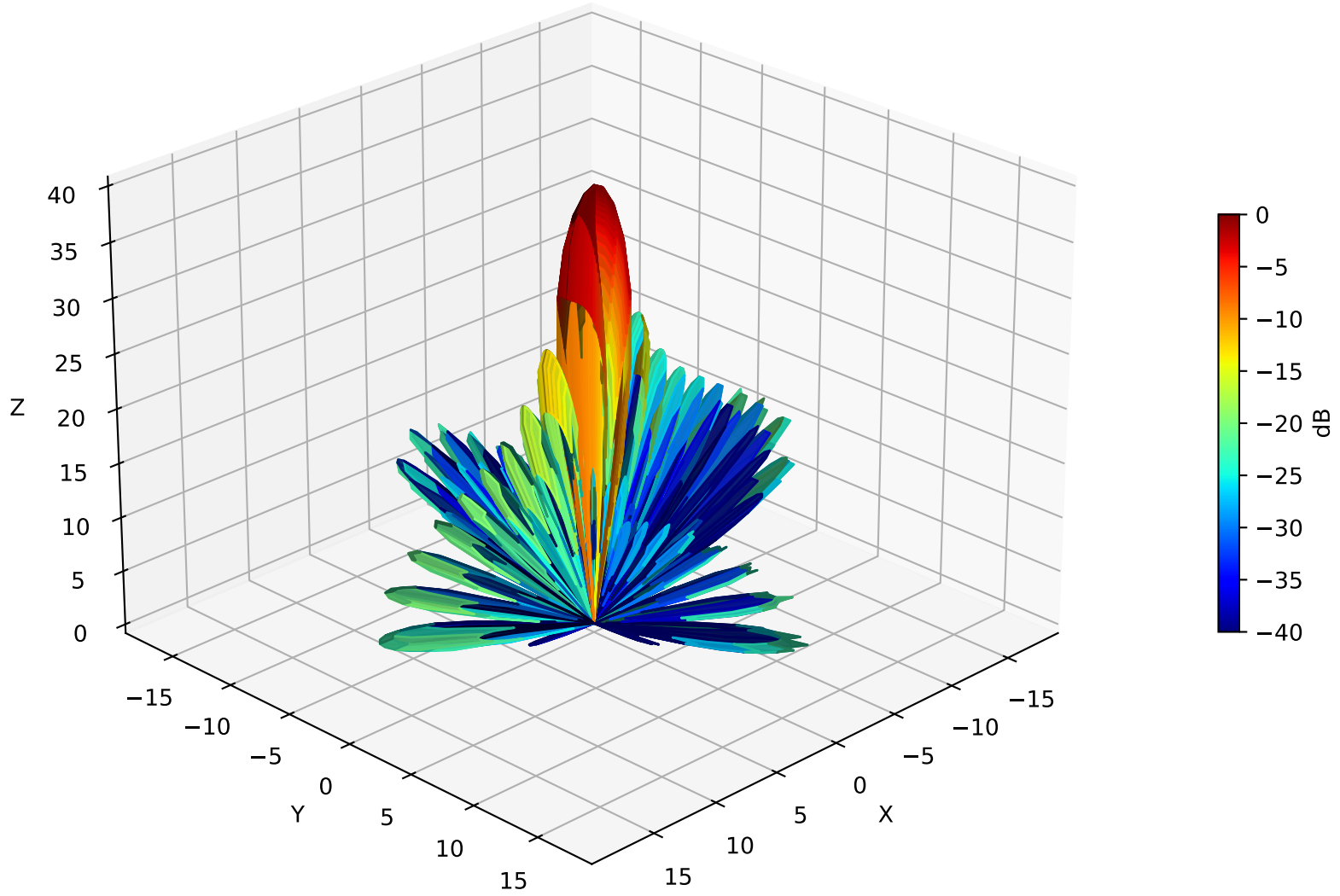
Radiation Pattern — 256 Elements at 10.5 GHz

Broadside beam · very narrow mainlobe



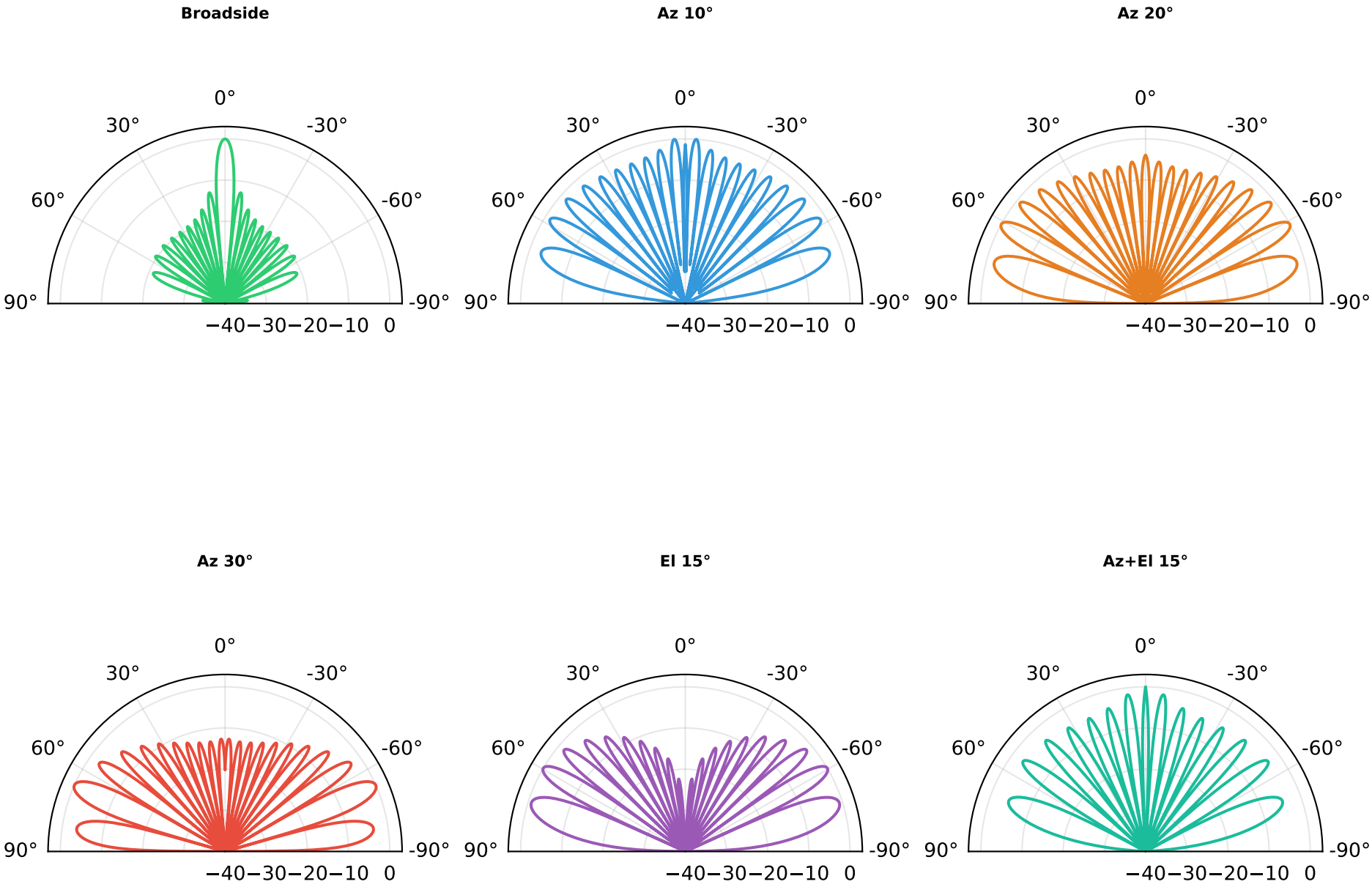
3D Radiation Pattern (upper hemisphere)

256-element array · pencil beam



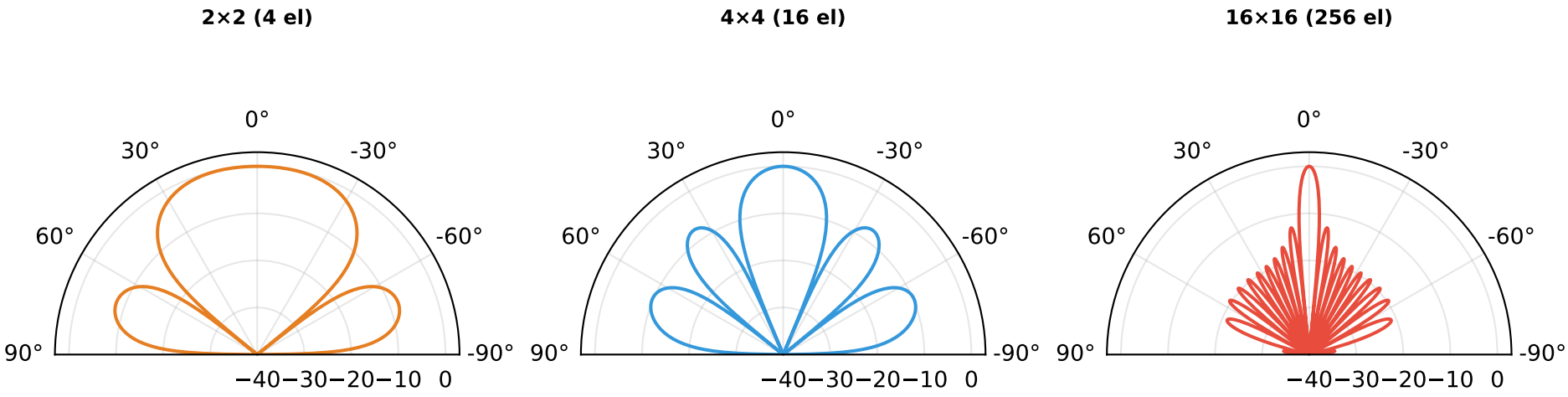
Electronic Beam Steering (256 elements)

Phase gradients applied across 16×16 grid · 64× ADAR1000



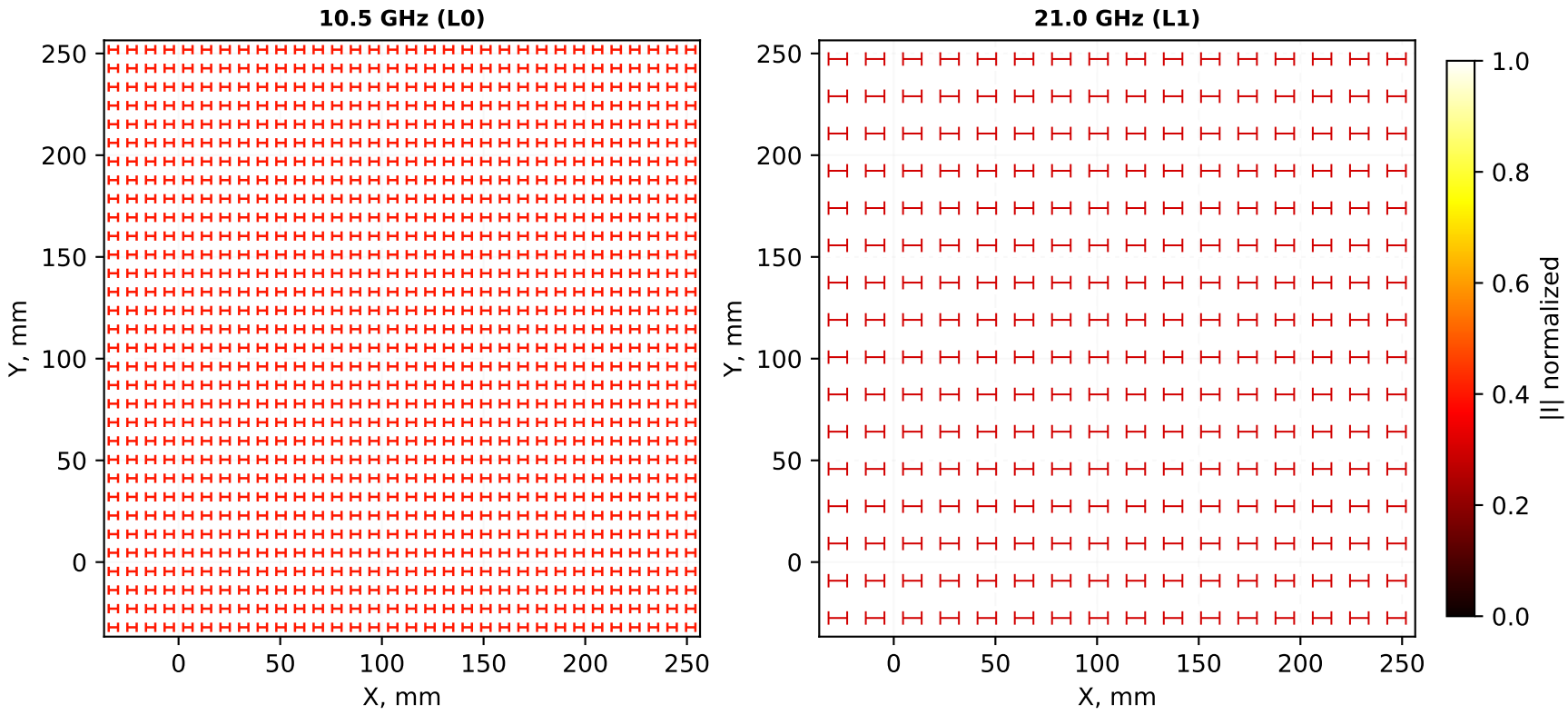
Array Size Comparison (E-plane broadside)

Beam narrows as aperture grows · same element pattern



Current Distribution — Full 256-Element Array

Color ∝ current magnitude per fractal level



Prototype Budget Estimate

All components at prototype quantities (2026 pricing)

System Budget & Cost Estimate

Component	Qty	Unit cost	Total
Tile PCB (73×110mm RO4350B)	16	$30	$480
ADAR1000 IC	64	$20	$1280
Backplane PCB (300×360mm RO4350B)	1	$250	$250
STM32F4 MCU	1	$8	$8
MMCX connectors	32	$2	$64
SPI cables + headers	16	$3	$48
Decoupling caps (100nF+10pF)	128	$0.05	$6
SMA connector	1	$5	$5
Power regulator module	1	$15	$15
Mechanical frame (Al)	1	$100	$100
Assembly + soldering	**1**	**$300**	**$300**
		TOTAL	$2,556

Simulation Results Summary

256-Element Tiled Phased Array · 10.5 GHz Radar

Parameter	Value	Unit
Design frequency	10.5	GHz
Elements	256 (16×16)	
Tiles	16 (4×4 grid)	
Beamformers	64× ADAR1000	
S_{11} @ f_0	-18.3	dB
VSWR	1.28	
Z_in (element)	49.3+j12.2	Ω
-10dB BW	1015	MHz
Directivity	33.0	dBi
Rad. efficiency	60	%
Feed network loss	~3.5	dB
Realized gain	27.3	dBi
HPBW (est.)	~5.3	deg
Scan range	±30	deg
Aperture	275×275	mm
Tile pitch	73.27	mm
Board (tile)	73×110	mm
Board (backplane)	300×360	mm
Est. cost (prototype)	~$2,500	USD
L0 resonance	10.5	GHz
L1 resonance	21.0	GHz
L2 resonance	42.0	GHz